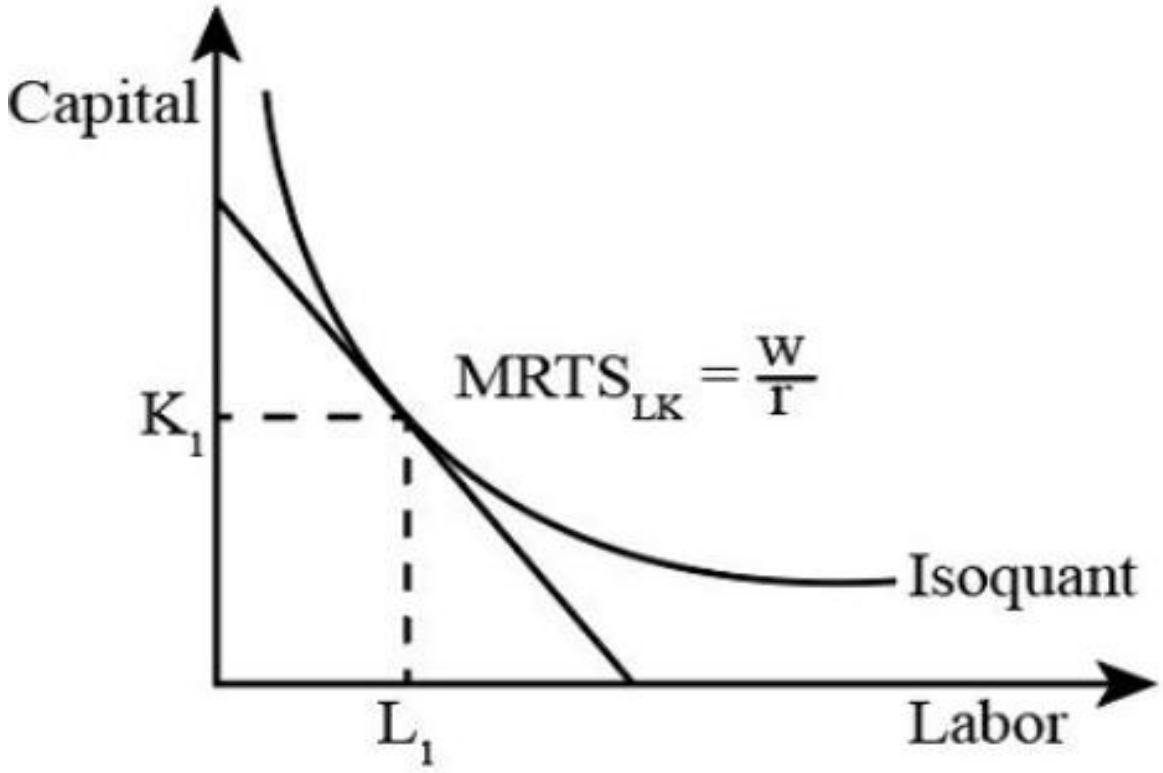
MICROECONOMICS

Isoquants and Isocost lines – Producer's Equilibrium



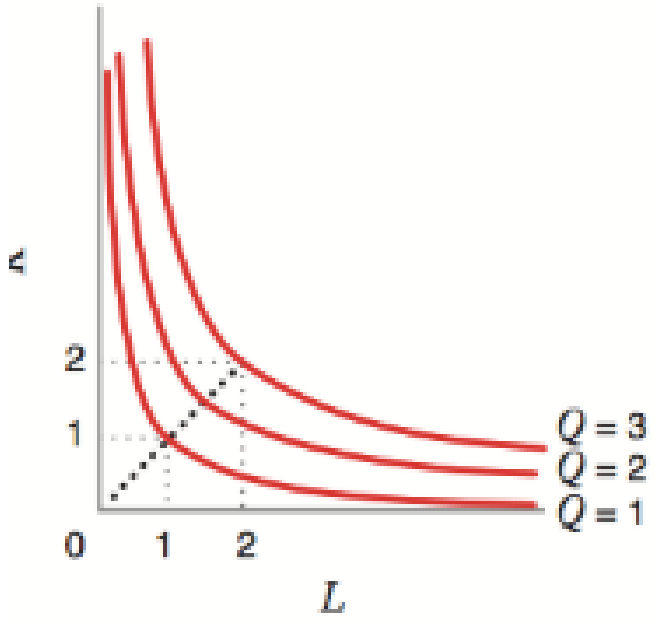
Returns to Scale

- When the inputs are increased in a certain proportion, if the output increases more than proportionately – Increasing returns

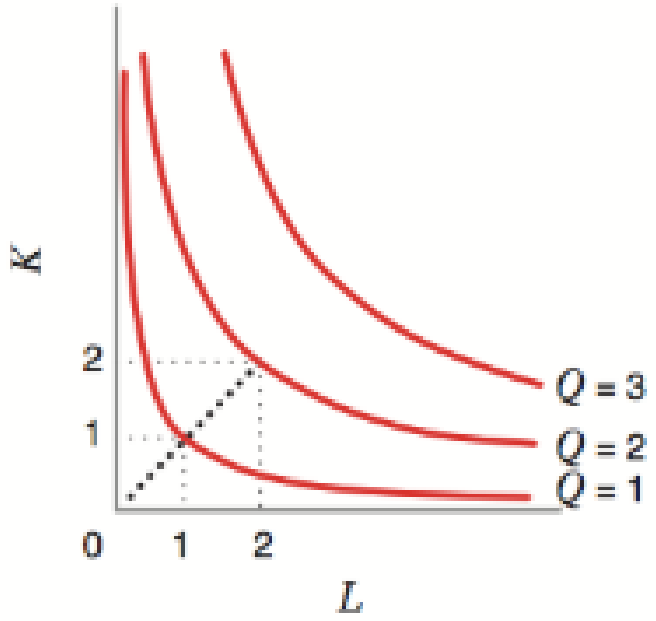
- When the inputs are increased in a certain proportion, if the output increases less than proportionately – Decreasing returns

- When the inputs are increased, if the output increases in the same proportion – Constant returns

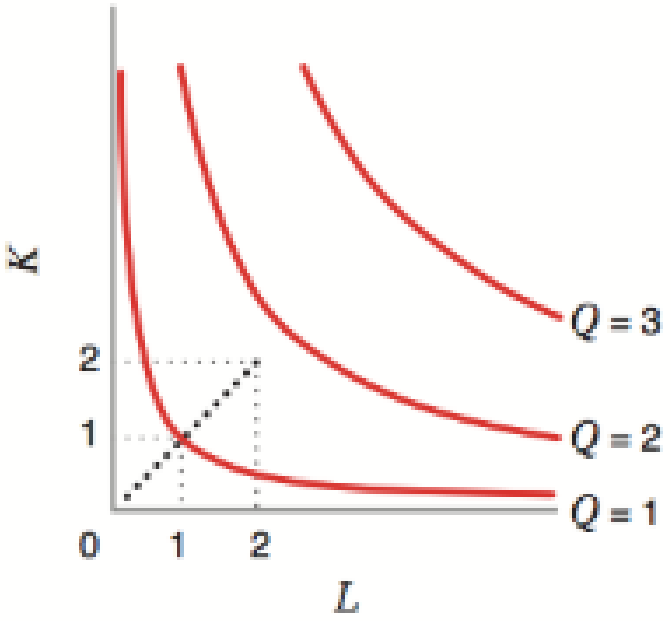
Returns to Scale



a) Increasing Returns to Scale



(b) Constant Returns to Scale



(c) Decreasing Returns to Scale

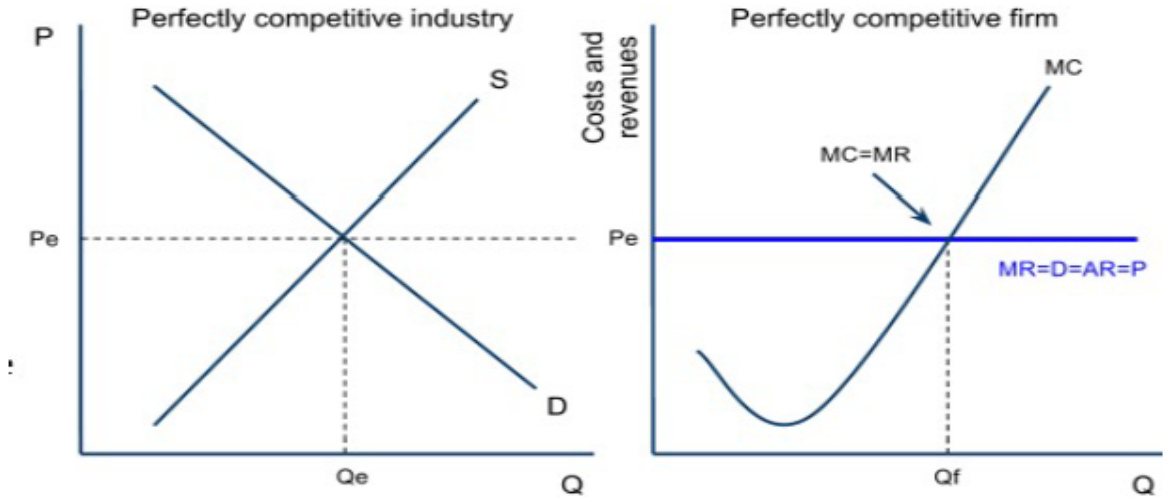
Perfect Competition

Characteristics :

- Large number of buyers and sellers
- Firm is price-taker
- Homogenous products
- Free entry and exit
- Perfect knowledge among buyers and sellers
- Perfect mobility of factors of production
- No additional selling costs

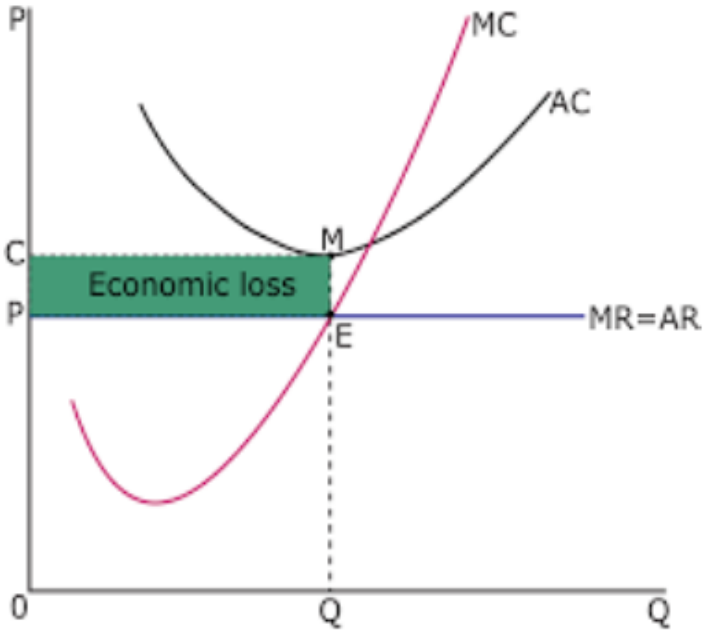
Perfect Competition

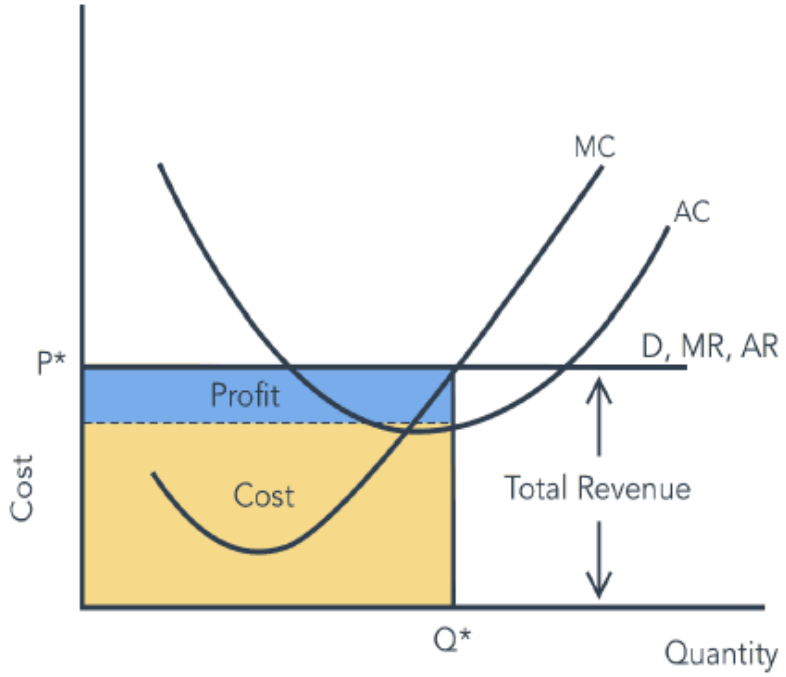
- The industry demand curve in the case of perfect competition is downward sloping and obeys the law of demand.
- The demand curve in case of a firm in perfect competition is a horizontal straight line parallel to the x-axis. It is given by $\boldsymbol{MR = D = AR = P}$
- The MR and AR earned on each unit is same and equal to P as price is given and constant. The horizontal demand curve indicates that at the given price the demand for good is perfectly elastic.
- **Equilibrium price and quantity is determined by the point where MC cuts MR from below** P=MC



Abnormal Profits

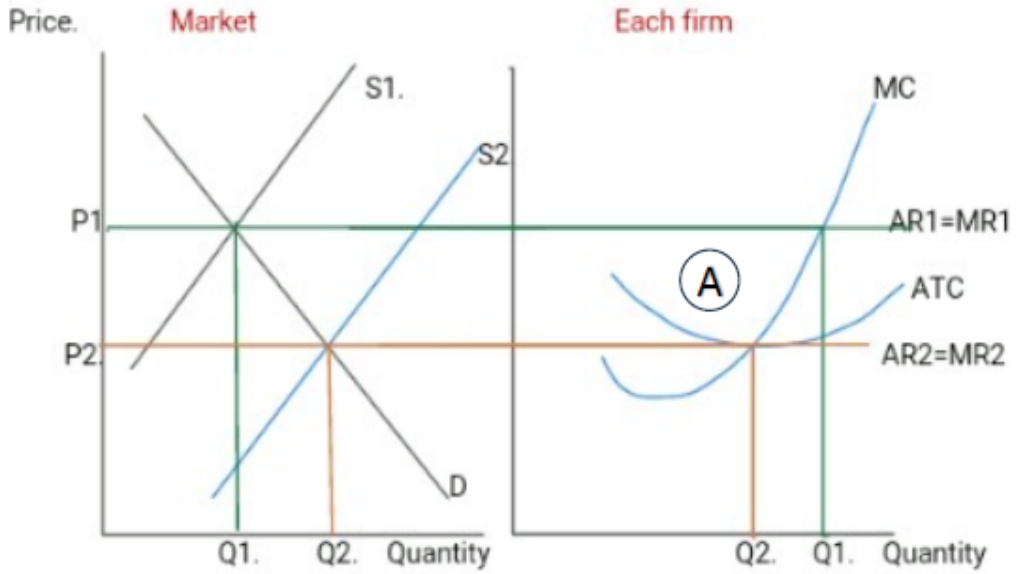
- In the short-run the firm might get abnormal profits or even loses, depending upon whether the AC is below or above the competitive price P.
- In the first figure, the firm suffers loss as the equilibrium price line is below the Average Cost (AC) curve. In the second figure, the firm earns profits as the equilibrium price is above the Average Cost (AC) curve.





Normal Profits

- **In the long-run the profits will be normal or zero economic profits. WHY?** If there are abnormal profits for a firm, lured by the profits, new firms enter the market and the quantity supplied shall increase pushing down the market price and hence driving out the abnormal profits.
- In the first figure, equilibrium price is P1 (D=S1). This price is above the Average Total Cost (ATC) and hence the firm earns super normal profits equal to rectangle A (second figure). But this abnormal profits attract many sellers into the market and as a result, the supply curve shifts to right and the new equilibrium price is P2 (D=S2). The price line is tangent to the minimum point of ATC and hence the firm gets normal profits or zero economic profit (second figure).



Monopoly

Main Features

- Single seller and several buyers
- Firm is a price-maker
- The product has no close substitutes
- Strong barriers to entry
- High market power
- Full control over supply

Why do monopolies arise?

- Exclusive ownership of key resources
- Exclusive rights of production given by the Government
- Extremely high fixed costs leading to natural monopolies

Monopolist's Revenue and Cost Curves

- AR=D is downward sloping as the firm/industry has to reduce the price of the good if it has to sell more units.
- The MR curve is downward sloping and falls more than the AR curve. For each additional unit sold; the firm receives the price of the unit sold (P), but at the same time it looses out the extra amount that it had received for each of the previous units sold.

Markup and Lerner's Index

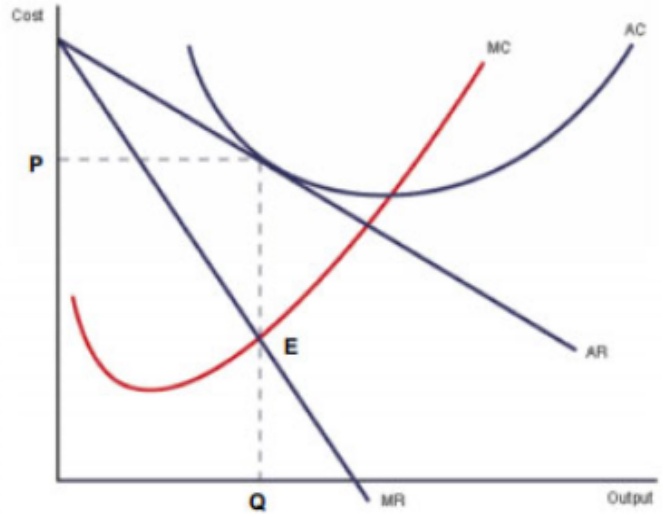
At equilibrium MR=MC, thus, $MC = P(1 + \frac{1}{E_d})$

$$\frac{\boldsymbol{P-MC}}{\boldsymbol{P}} = -\frac{\boldsymbol{1}}{\boldsymbol{E_d}}$$

- $\frac{\boldsymbol{P-MC}}{\boldsymbol{P}}$ **is the markup over marginal cost as a percentage of price** [The Rule of Thumb of pricing in monopoly says that this markup should be equal to negative inverse of elasticity of demand]
- $\frac{P-MC}{P}$, also known as the Lerner's index determines the power of monopoly.
- The monopoly power also depends on the number of firms, interaction between the firms and the elasticity of demand of the market.

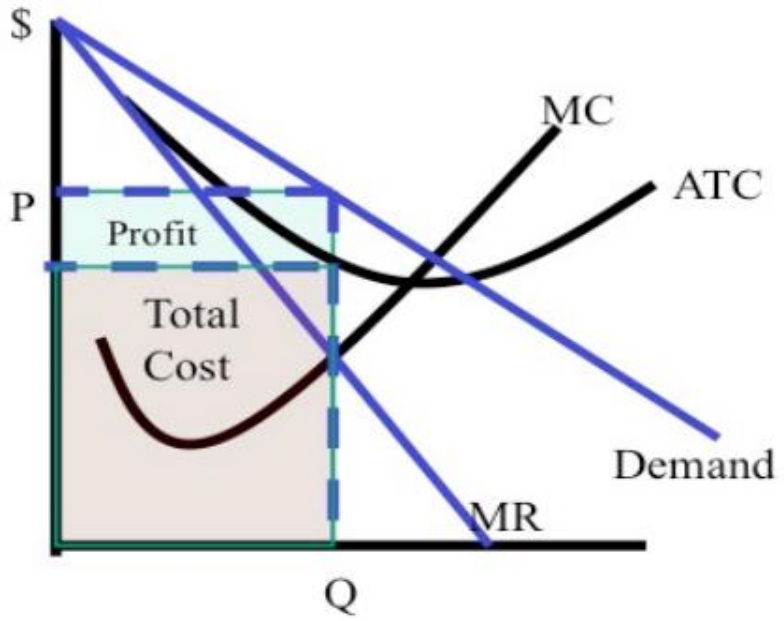
Monopolist's Equilibrium

- $MR = MC$ is the profit maximising condition and hence equilibrium is obtained at point E.
- Q is the equilibrium quantity and P is the equilibrium price which is higher than the equilibrium price of the perfectly competitive market.
- If the firm produces an output less than Q, then $MC < MR$ and the firm will produce more till it reaches $MR = MC$. Similarly, if an output more than Q is produced, then $MC > MR$ which will urge the firm to produce lesser and be more efficient.



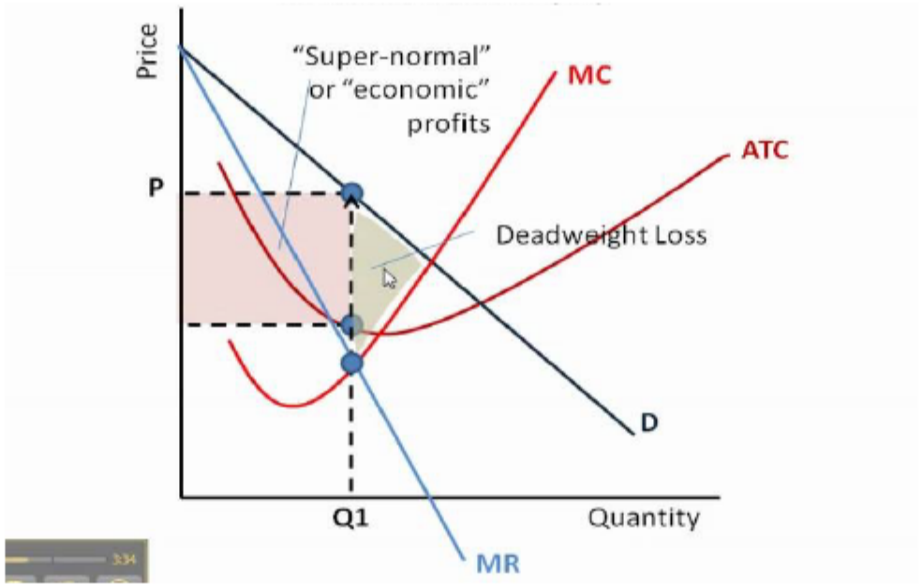
Monopolist's Profits

- A monopolist can obtain extremely high profits depending upon its power of monopoly.
- The area above the the Average Total Cost (ATC) curve and the equilibrium price is deemed as the firm's profits.
- If the demand is relatively inelastic, a monopolist will be able to charge high prices for the product.

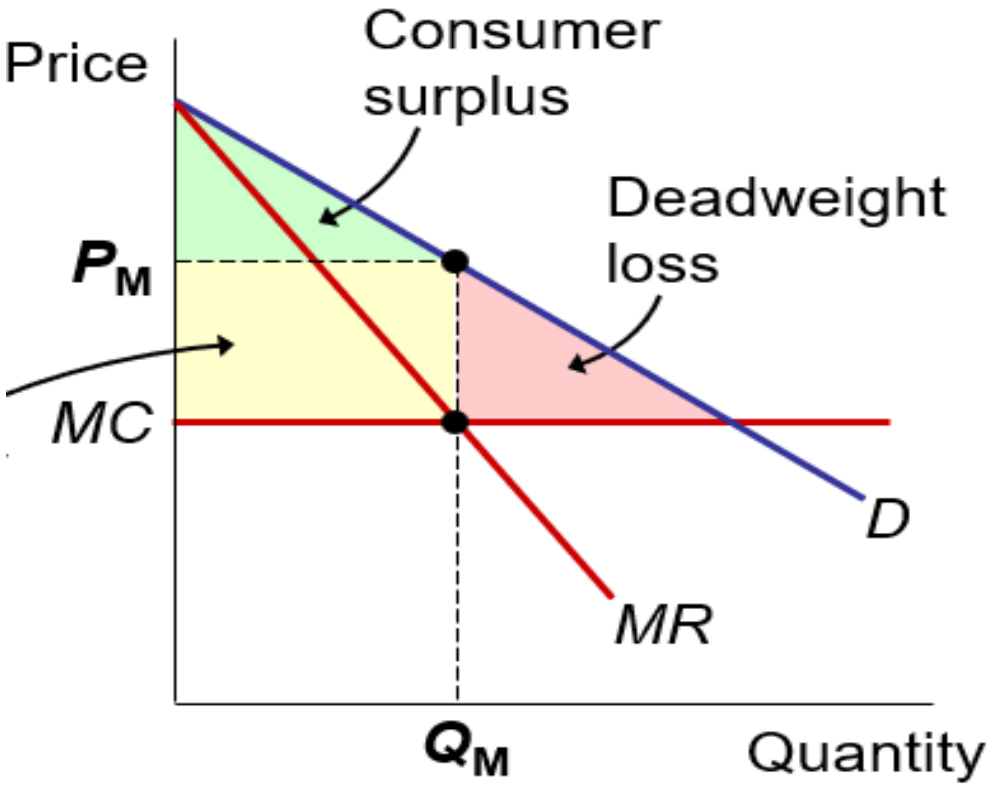


Loss of Welfare

- The equilibrium output obtained in the case of perfect competition (D=MC) will always be higher than the equilibrium output in the case of a monopoly (MC=MR).
- This reduced output takes away a part of consumer surplus and producer surplus which then becomes "deadweight loss".



DWL and Consumer Surplus



Monopolistic Competition

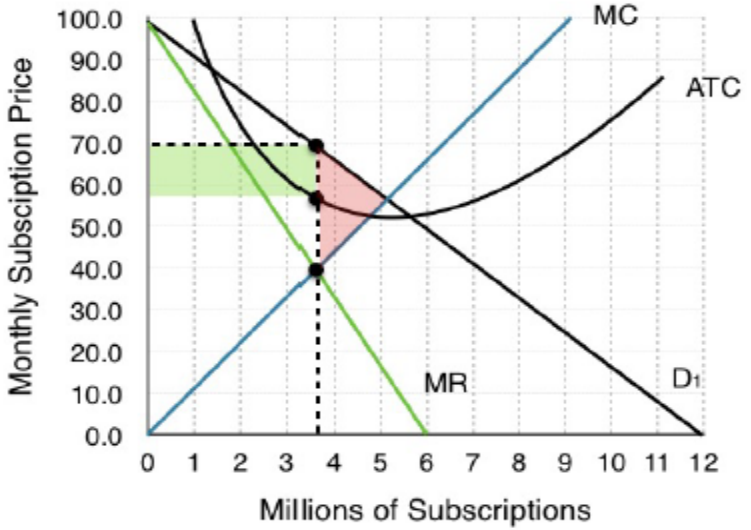
Characteristics:

- Many buyers and sellers
- Differentiated products
- Freedom of entry and exit
- Independent decision making and some amount of control over the price for sellers
- Advertising and other selling costs
- More elastic demand
- Incomplete mobility of products
- Imperfect knowledge

Monopolistic Competition

- Equilibrium happens at the point where $MR = MC$. The equilibrium price is at 70.
- The price in the case of a monopolistic firm will not be as high as in the case of a monopoly.
- In the figure, the green shaded area is the profits (above ATC and below equilibrium price) and the orange shaded area is the deadweight loss.

Is deadweight loss/loss of welfare lesser in monopolistic competition compared to monopoly? If so, why?



Oligopoly

Characteristics

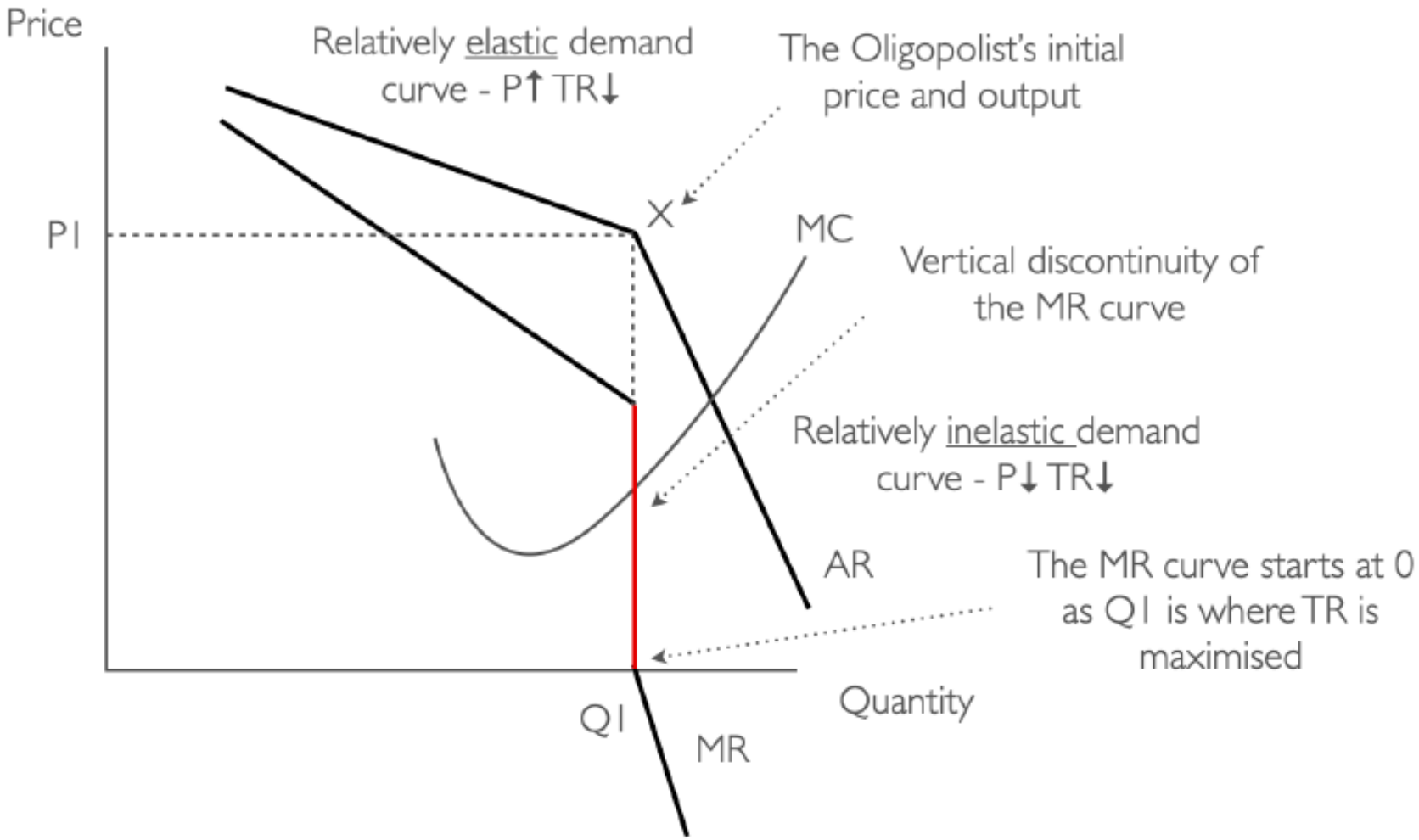
- Few firms
- Non-price competition
- Interdependence among the firms
- High barriers to entry
- Differentiated or homogenous products
- Group behaviour in setting price

Oligopoly : Kinked Demand Model

- The demand curve ($D = AR$) in the case of oligopoly is kinked at the prevailing price P1 of the oligopolist. The point of kink is given as X in the figure.
- If one firm increases the price above this point, other firms might not respond by raising prices and thus the loss in the firm's market share and revenue will be larger. Above the point of kink, the elasticity is high because quantity demanded is highly responsive to price in this area.
- Similarly, if the firm reduces the price below the point of kink, other firms might respond by lowering the prices which results in the firm not gaining much profits as it expected. Thus, in this area the the gain in quantity demanded by lowering the price is lower and hence the demand curve is relatively inelastic.
- The corresponding MR curve will have a discontinuity at the point of kink. If the MC curve passes through the region of discontinuity, then the price and quantity shall remain unchanged.

Why does price rigidity exists in oligopoly?

Oligopoly Equilibrium



THANK YOU